TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature's low-rank (dipolar) component, on *which* lead, is even recoverable. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S, and a target lead ℓ, two closed-form numbers derived from one truncated SVD of the segment's empirical rank-3 spatial ("dipolar") sub-matrix—an *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target's low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$ (its noise/observed-residual gain). The per-lead criterion is the classical estimability condition specialized to ECG (exact vs. a numerical ϱ-truncation, which we separate). The robust findings: precordial ST is *exactly* non-identifiable from limb leads (the limb leads span the frontal plane, so $\boldsymbol{M}_{s,S}$ has exact rank 2), and the graded normalized-identifiability ordering is bootstrap-stable for QRS. For the low-amplitude ST/T segments the fine ordering is *sensitive to the limb-lead weighting* (a duplicated-limb sensitivity analysis), so we report it with that caveat rather than as an exact graded claim. We attach distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression, strict fold discipline, honestly stated within-group coverage), and observe that the total ST-threshold error rate is empirically similar across the evaluated reconstructors (not a certified bound) while the false-positive/false-negative split is a reconstructor choice. The map turns "trust the reconstruction" into a per-feature, per-lead, calibrated statement.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead's dipolar morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment's instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis $\boldsymbol{M}_s$ (top-3 singular vectors of the population lead covariance). We stress that $\boldsymbol{M}_s$ is a *population-estimated (PCA)* object: on real PTB-XL its three principal angles to the classical vectorcardiographic (inverse-Dower) space are graded—one close (2–6°), one moderate (10–15°), and one substantially different (43–55°)—so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 4).

Related work and contributions. Classical work reconstructs missing leads (regression / VCG synthesis [3, 4, 5], deep and arbitrary-mask waveform-completion models [1, 2, 6, 7]) or *selects* optimal leads [8]; whether a given target is even recoverable is the classical estimability question [9]. Those systems produce a reconstruction; we instead certify, before any reconstruction, which target is recoverable and attach a graded, per-target-lead recoverability measure with calibrated per-record error. Our neural baseline is a *representative* masked-completion U-Net (same family as [1, 6]), not a state-of-the-art claim. We do not claim the underlying identity as new. Our contributions are: (1) a *graded*, per-target-lead recoverability map—continuous $\eta_{s,\ell}(S)/\kappa_{s,\ell}(S)$ from a single truncated SVD that *rank* target leads (not a binary rank test), over an *empirically-estimated* $\boldsymbol{M}_s$ that differs 43–55° from the textbook inverse-Dower dipole, with a truncation-tolerance sensitivity analysis and record-bootstrap CIs (Sec. 2); (2) a calibration layer attaching distribution-free intervals (a sound application of conformalized quantile regression + Mondrian [10], not a new method) to the map per feature and lead, with strict fold discipline and honest within-group (not per-diagnosis) coverage (Sec. 3); (3) a graded ST-safety study on PTB-XL—continuous full-waveform reconstruction shows anterior precordial ST is far from recoverable from limb leads, with a total ST-threshold error rate empirically similar across the evaluated reconstructors (an empirical observation, not a certified bound) and a false-positive/false-negative split that is a reconstructor choice (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Model. Within segment s write the population 12-lead vector as $\boldsymbol{L}_s = \boldsymbol{\mu}_s + \boldsymbol{M}_s\boldsymbol{d} + \boldsymbol{r}_s$, where $\boldsymbol{M}_s \in \mathbb{R}^{12\times 3}$ is the (population-estimated, PCA) dipolar basis, $\boldsymbol{d}$ the dipole coordinates, and $\boldsymbol{r}_s$ the non-dipolar residual. Observing S gives $\boldsymbol{y}_S = \mathbf{Sel}_S\boldsymbol{L}_s + \boldsymbol{n}$, with $\boldsymbol{M}_{s,S} = \mathbf{Sel}_S\boldsymbol{M}_s$. Write the exact Moore–Penrose pseudo-inverse $\boldsymbol{M}_{s,S}^{+}$ and the exact row-space projector $\boldsymbol{P}_{\rm obs} = \boldsymbol{M}_{s,S}^{+}\boldsymbol{M}_{s,S}$. For numerical deployment we also use the *rank-r truncation* at relative tolerance ϱ, $\boldsymbol{M}_{s,S}^{(\varrho)} = \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^{\top}$ keeping singular triplets with $\sigma_i > \varrho\,\sigma_{\max}$, with pseudo-inverse $\boldsymbol{M}_{s,S}^{(\varrho)+}$ and projector $\boldsymbol{P}_{\rm obs}^{(\varrho)} = \boldsymbol{V}_r\boldsymbol{V}_r^{\top}$.

Proposition 1 (Exact per-target-lead dipolar identifiability and conditioning)**.** *For each target lead ℓ define $\eta_{s,\ell}(S) = \left\|\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s(\boldsymbol{I} - \boldsymbol{P}_{\rm obs})\right\|_2$ and $\kappa_{s,\ell}(S) = \left\|\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}\right\|_2$ from the* exact $\boldsymbol{M}_{s,S}$.

- *(i)* Identifiability. $\eta_{s,\ell}(S) = 0$ *iff $\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s$ lies in the exact row space of $\boldsymbol{M}_{s,S}$; then $\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{d}$ is a deterministic function of $\boldsymbol{M}_{s,S}\boldsymbol{d}$—two dipoles with the same observation have the same lead-ℓ dipolar value, at any SNR. If $\eta_{s,\ell}(S) > 0$ there is a $\boldsymbol{\delta} \in \ker \boldsymbol{M}_{s,S}$ (an unobserved dipole direction) with $\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{\delta} \neq 0$; then $\boldsymbol{d}$ and $\boldsymbol{d} + t\boldsymbol{\delta}$ produce identical observations for all t while lead ℓ's dipolar value changes, so it is not identifiable at any SNR.*
- *(ii)* Conditioning. *On the identifiable part, the lead-ℓ error from observation noise and observed non-dipolar residual obeys $\left|\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}(\mathbf{Sel}_S\boldsymbol{r}_s + \boldsymbol{n})\right| \leq \kappa_{s,\ell}(S)\left(\|\mathbf{Sel}_S\boldsymbol{r}_s\|_2 + \|\boldsymbol{n}\|_2\right)$.*

The global $\kappa_s(S) = \|\boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}\|_2$ (spectral norm) upper-bounds every per-lead $\kappa_{s,\ell}(S)$ (a row norm never exceeds the spectral norm; it is not their maximum).

Exact vs. ϱ-identifiability (numerical deployment). The deployed map computes $\eta_{s,\ell}^{(\varrho)}, \kappa_{s,\ell}^{(\varrho)}$ from the truncation $\boldsymbol{P}_{\rm obs}^{(\varrho)}$, so a singular direction observed only through a value below $\varrho\,\sigma_{\max}$ is treated as unobserved. Then $\eta_{s,\ell}^{(\varrho)} > 0$ means lead ℓ is *not stably recoverable at resolution* ϱ (outside the retained numerical row space); it does *not* by itself imply the exact matrix has a null direction—a tiny-but-nonzero singular value is exactly identifiable ($\eta = 0$) yet ϱ-unidentifiable ($\eta^{(\varrho)} > 0$), as we verify numerically (`tests/test_exact_vs_rho.py`). We therefore call the $\eta^{(\varrho)}$ heatmap the *ϱ-identifiability* map; for well-separated configurations the exact and ϱ verdicts coincide, and near-rank-deficient S are reported with a ϱ-sweep and record-bootstrap CIs. **The limb-6 flagship is exact:** the six limb leads are exact linear combinations of I and II (Einthoven/Goldberger), so $\boldsymbol{M}_{s,S}$ has exact rank 2 and precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ. This exactness is conditional on the a-priori rank-3 dipole model for $\boldsymbol{M}_s$: at retained rank 2 the limb-6 observation spans all of $\boldsymbol{M}_s$ and the precordial verdict collapses to η=0; the non-identifiability appears at rank $\geq$3 (the third dipolar direction), so "exact" means exact *given* rank 3.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [9]) for the linear functional $\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{d}$; we claim no novelty for this identity. The contribution (Sec. 4) is its use as a graded, empirically-estimated recoverability map with per-record calibrated error rates (Sec. 3).

The non-dipolar residual (no independence claim on real data). Let $\boldsymbol{m}_s(\boldsymbol{y}_S) = \mathbb{E}[\boldsymbol{r}_s \mid \boldsymbol{y}_S]$ and $\boldsymbol{u}_s = \boldsymbol{r}_s - \boldsymbol{m}_s(\boldsymbol{y}_S)$, so $\mathbb{E}[\boldsymbol{u}_s \mid \boldsymbol{y}_S] = \boldsymbol{0}$. We do *not* assume $\boldsymbol{u}_s \perp \boldsymbol{y}_S$. Under squared loss the Bayes-optimal reconstruction error of the (vector) residual is

$$\inf_{\hat{\boldsymbol{r}}} \mathbb{E}\left\|\boldsymbol{r}_s - \hat{\boldsymbol{r}}(\boldsymbol{y}_S)\right\|_2^2 = \mathbb{E}\left[\operatorname{tr}\operatorname{Cov}(\boldsymbol{r}_s \mid \boldsymbol{y}_S)\right].$$

Part of $\boldsymbol{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\boldsymbol{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—*not* declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only)**.** *If the data are generated so that a component $\boldsymbol{u}$ is statistically independent of $\boldsymbol{y}_S$ (an explicit synthetic construction), then $p(\boldsymbol{u} \mid \boldsymbol{y}_S) = p(\boldsymbol{u})$, the Bayes estimate of $\boldsymbol{u}$ under squared loss is its prior mean, and $\mathbb{E}\|\widehat{\boldsymbol{u}} - \boldsymbol{u}\|_2^2 \geq \operatorname{tr}\operatorname{Cov}(\boldsymbol{u})$ for any estimator. This strict limit is invoked* only *for the synthetic model; on real ECG we make no such independence claim.*

3. DISTRIBUTION-FREE CALIBRATION

3.1. Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment's samples). We train a gradient-boosted quantile regressor of it on the observed leads' segment means and apply Mondrian conformalized quantile regression [10] per (S, s, ℓ) group, with strict fold discipline: basis and model on folds 1–7, the regressor's `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal $1-\alpha$=0.90). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.87–0.94; e.g. {I,II,V1,V3,V5} QRS/V2 0.91, record-bootstrap CI $[0.89, 0.93]$, mean width $0.150\,\mathrm{mV}$).

What is and is not guaranteed. Mondrian CQR gives *within-*(S, s, ℓ)*-group marginal* coverage under exchangeability—*not* coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 (n=85), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

4. EXPERIMENTS

Data. PTB-XL [11] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [12].

Reproducibility. Code, fixed and deterministic seeds, a pinned dependency manifest, result lineage (commit, a working-tree-clean flag, and dataset/split hashes in every JSON), and a one-command pipeline are released[1]; every number in the tables is regenerated from result JSON (no hand-typed values). The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, Adam 2×10^{-4}, MSE on masked leads under random masks, 3 training seeds); the residual predictor is a gradient-boosted quantile regressor. The map is a deterministic function of the estimated $\boldsymbol{\mu}_s, \boldsymbol{M}_s$ (record-level bootstrap CIs), not a held-out evaluation.

4.1. The recoverability map on PTB-XL

We estimate $\boldsymbol{M}_s$ (rank 3; the cardiac dipole is 3-D *a priori*, and the cumulative explained variance confirms rank 3 captures the bulk—a rank-sensitivity sweep r=2 . . . 5 is reported in `results/recoverability_maps.json`) per segment on the training NORM records, with RECORD-LEVEL bootstrap CIs (resample records, refit $\boldsymbol{M}_s$; ϱ=10^{-2}; Fig. 1). We plot the *normalized* identifiability $\tilde{\eta}_{s,\ell} = \eta_{s,\ell}/\|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s\|_2$—a dimensionless relative unobserved geometric gain (a ratio of L_2 norms, not a variance fraction). A single observed lead leaves every other lead unidentifiable; a dipole-spanning {I,II,V2} or {I,II,V1,V3,V5} makes all targets *exactly* identifiable (η=0).

The robust flagship is exact. The six limb leads are exact linear combinations of I and II, so $\boldsymbol{M}_{s,S}$ for limb-6 has *exact* rank 2: precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ (Prop. 1).

Graded ordering: stable for QRS, weighting-sensitive for ST/T. A duplicated-limb sensitivity analysis (`results/leadweight.json`; refit on the 8 algebraically independent leads, on the *same*

[1] Anonymized for review; released on acceptance. Camera-ready artifacts are regenerated from a clean tree (the lineage `git_dirty` flag is asserted false).

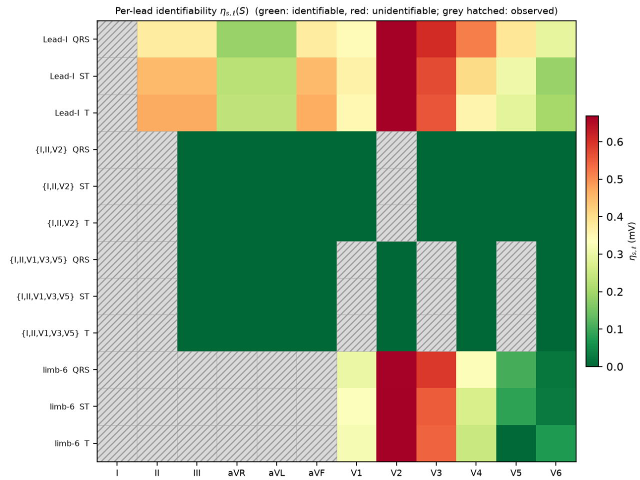

Fig. 1. Recoverability map: normalized identifiability $\tilde{\eta}_{s,\ell}(S)$ (left, dimensionless) and prior-conditional expected ambiguity in mV (right), colorblind-safe `cividis`; grey hatched = observed. For limb-6 all precordial leads are unidentifiable (η>0); the fine ST/T ordering is weighting-sensitive (text).

records) shows the V1–V6 $\tilde{\eta}$ ordering is bootstrap-stable for QRS (Spearman 1.00, CI lower bound 0.88; anterior-before-lateral in 0.98 of resamples) but *not* robust for the low-amplitude ST/T segments (ST Spearman point 0.89 with a wide bootstrap CI; anterior-before-lateral in only 0.77 of resamples). We therefore report the ST/T $\tilde{\eta}$/ambiguity magnitudes with this caveat and *do not* claim a fine ST/T ordering; the binary limb→precordial verdict above is the robust ST statement. The estimated subspace also depends on diagnosis (a NORM vs. MI principal-angle sensitivity is reported), so the map is a *NORM-trained reference*.

Downstream ST cost across reconstructors. We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads exact; ST at J+60 ms vs. the isoelectric PR-segment baseline [P offset − QRS onset], on fiducials from the observed Lead II shared between truth and reconstruction; median over beats; the PR-segment baseline fell back to the pre-QRS window in ≈ 14% of beats), and count ST-threshold events (|ST|≥0.1 mV; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 1). On 1498 common records the *total* wrong-event rate is empirically similar across reconstructors (51.1–53.7%) and is *dominated by false negatives* (missed true crossings: limb→precordial recovery smooths ST toward the population), with the false-positive/false-negative split a reconstructor choice (paired record-bootstrap Δ significant). We do not derive a minimax/Bayes bound, so we call the shared total an empirically similar total error rate, *not* a certified lower bound; we report ST-threshold events, not diagnoses.

Truncation-tolerance sensitivity. Well-posed config-

Table 1. Limb-6 $\to$ precordial (all V1–V6) ST reconstruction, 1498 common fold-10 records. FP/FN = false-positive/false-negative ST-threshold crossings (%); total = FP+FN. FN dominates (missed crossings); the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	|ST| err (mV)
dipolar	5.3	45.7	51.1	0.077
ridge	1.7	52.1	53.7	0.076
OLS	1.7	51.8	53.5	0.076

urations are ϱ-stable ({I,II,V1,V3,V5} is rank 3 across $\varrho \in \{10^{-4}, \ldots, 10^{-1}\}$); near-rank -deficient sets are ϱ-sensitive and reported with the ϱ-sweep and bootstrap CIs. A tiny-but-nonzero singular value is exactly identifiable yet ϱ-unidentifiable, which is why we separate exact from ϱ-identifiability (Prop. 1).

4.2. Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge λ selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead masks (one model serves any configuration, no target-lead leakage), the same family as [1, 6, 7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge$\to$U-Net step is within its paired CI)—prior-mean $\to$ dipolar (Tier I) $\to$ per-segment ridge (Tier I+II) $\to$ U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model's margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on {I,II,V1,V3,V5}; paired Δ CI $[-0.007, +0.000]$ mV). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS $0.481 \to 0.238$)—it recovers the predictable, largely population-correlated structure—but plateaus at $1.46\times$ its spanning-set error (paired spanning$\to$limb-6 Δ CI $[+0.070, +0.079]$ mV) and no reconstructor closes that gap. This residual gap is *consistent with* the unobserved dipolar coordinate the map's $\eta>0$ marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equality): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket "not recoverable."

Empirical, not physical, subspace. On real PTB-XL the estimated $\boldsymbol{M}_s$ has three graded principal angles to the classical inverse-Dower vectorcardiographic space [3, 4]: one close (2–6°), one moderate (10–15°), and one substantially different (43–55°, bootstrap CIs excluding small angles). We therefore treat $\boldsymbol{M}_s$ as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

Table 2. Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). "dipolar" = Tier I; "ridge" = per-segment ridge (Tier I+II); "U-Net" = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.467	0.236	0.220	**0.183**
{I,II,V1,V3,V5}	0.481	0.183	0.167	**0.164**
limb-6	0.481	0.361	0.344	**0.238**

5. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a graded per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It states which lead configurations support recovery of a feature's low-rank component, e.g. that precordial ST is exactly non-identifiable from limb leads, and reports the empirically similar total ST-threshold error rate the evaluated reconstructors incur there (dominated by missed, not fabricated, crossings).

6. REFERENCES

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